SERPINE1 promoter-reporter architectures

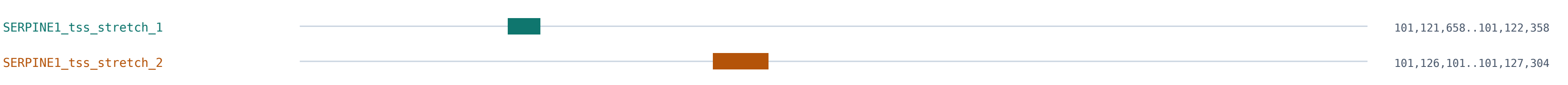
`5' -> 3' display, genomic 101117158 -> 101140263 | strand + | assembly GRCh38 | annotation Ensembl live GRCh38 | upstream 5000 bp | downstream 1000 bp`

Canonical reporter comparison: 3 TSS class(es), 3 transcript(s), 9 explicit architecture row(s); normalized evidence remains observational/predictive context.

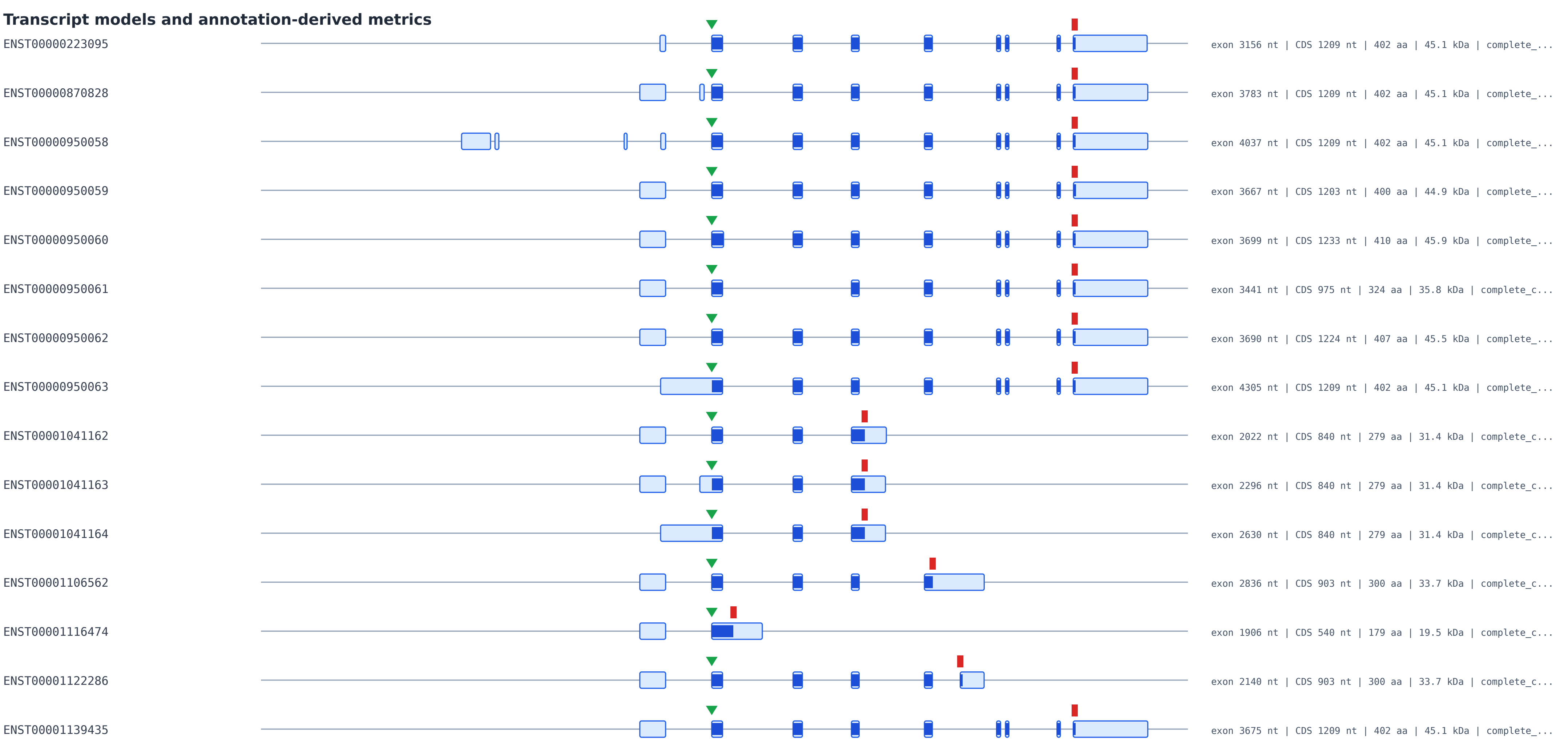
Legend (annotation-derived transcript features)

- exon (light blue); intron (line)
- coding sequence (CDS)
- annotated start codon
- annotated stop codon

TSS stretches: reference for similarity below



Transcript models and annotation-derived metrics

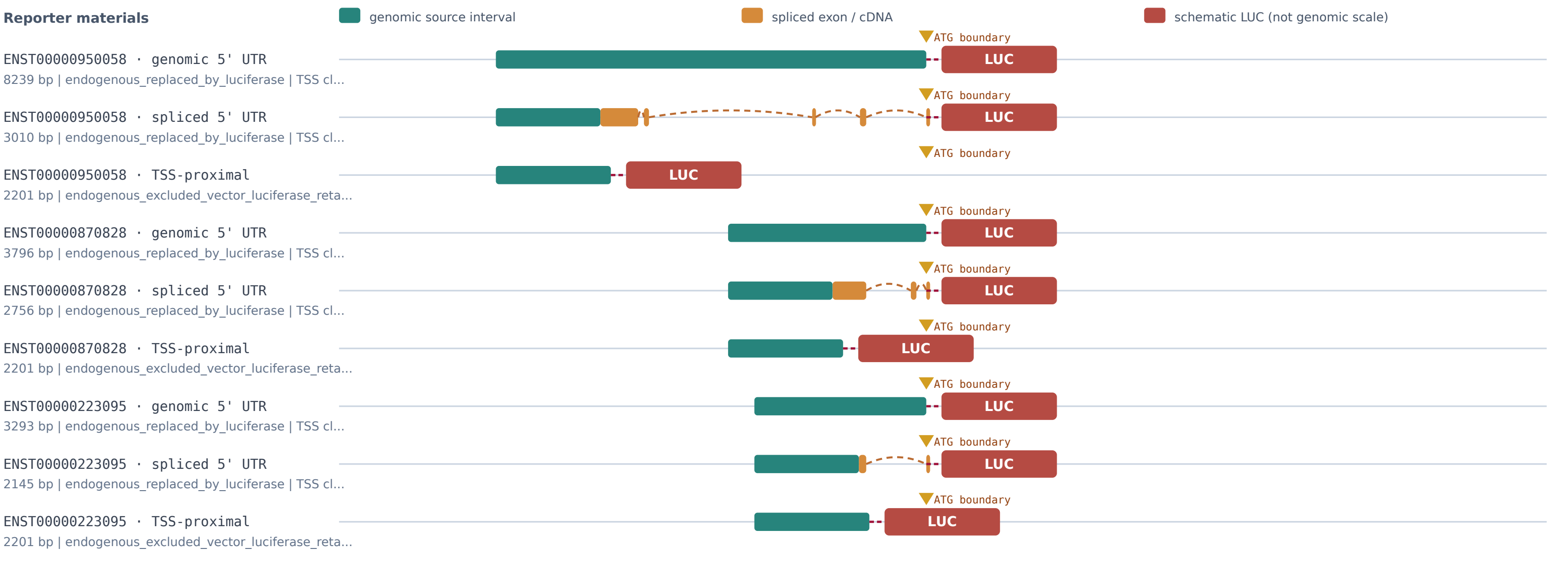

ENST00000223095
exon 3156 nt | CDS 1209 nt | 402 aa | 45.1 kDa | complete_...
ENST00000870828
exon 3783 nt | CDS 1209 nt | 402 aa | 45.1 kDa | complete_...
ENST00000950058
exon 4037 nt | CDS 1209 nt | 402 aa | 45.1 kDa | complete_...
ENST00000950059
exon 3667 nt | CDS 1203 nt | 400 aa | 44.9 kDa | complete_...
ENST00000950060
exon 3699 nt | CDS 1233 nt | 410 aa | 45.9 kDa | complete_...
ENST00000950061
exon 3441 nt | CDS 975 nt | 324 aa | 35.8 kDa | complete_c...
ENST00000950062
exon 3690 nt | CDS 1224 nt | 407 aa | 45.5 kDa | complete_...
ENST00000950063
exon 4305 nt | CDS 1209 nt | 402 aa | 45.1 kDa | complete_...
ENST00001041162
exon 2022 nt | CDS 840 nt | 279 aa | 31.4 kDa | complete_c...
ENST00001041163
exon 2296 nt | CDS 840 nt | 279 aa | 31.4 kDa | complete_c...
ENST00001041164
exon 2630 nt | CDS 840 nt | 279 aa | 31.4 kDa | complete_c...
ENST00001106562
exon 2836 nt | CDS 903 nt | 300 aa | 33.7 kDa | complete_c...
ENST00001116474
exon 1906 nt | CDS 540 nt | 179 aa | 19.5 kDa | complete_c...
ENST00001122286
exon 2140 nt | CDS 903 nt | 300 aa | 33.7 kDa | complete_c...
ENST00001139435
exon 3675 nt | CDS 1209 nt | 402 aa | 45.1 kDa | complete_...

Proposed reporter architectures

source intervals use the shared axis; synthetic tails are schematic

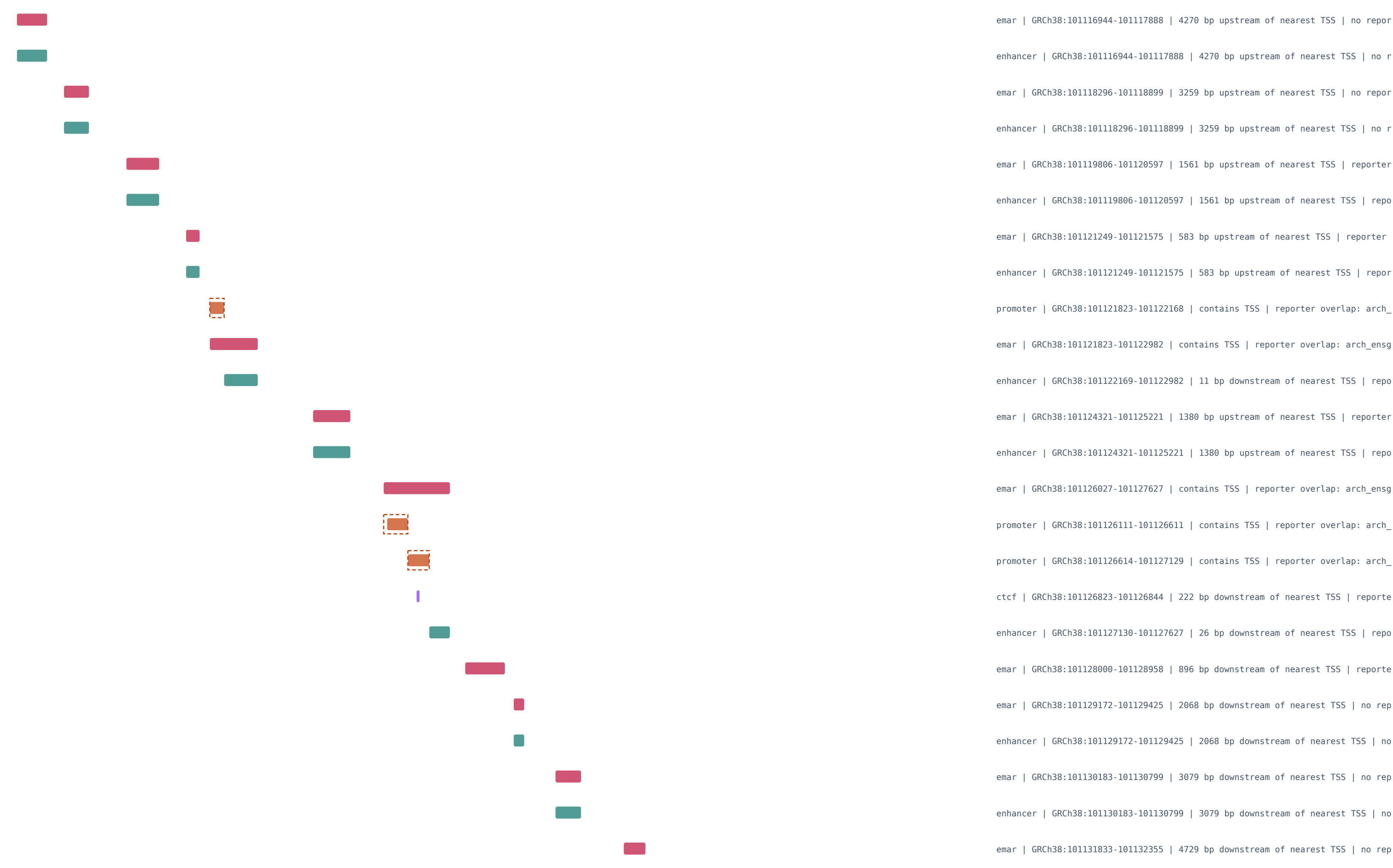
Reporter materials

- genomic source interval
- spliced exon / cDNA
- schematic LUC (not genomic scale)


ATG boundary
LUC
ENST00000950058 · genomic 5' UTR
8239 bp | endogenous_replaced_by_luciferase | TSS cl...
ENST00000950058 · spliced 5' UTR
3010 bp | endogenous_replaced_by_luciferase | TSS cl...
ENST00000950058 · TSS-proximal
2201 bp | endogenous_excluded_vector_luciferase_reta...
ENST00000870828 · genomic 5' UTR
3796 bp | endogenous_replaced_by_luciferase | TSS cl...
ENST00000870828 · spliced 5' UTR
2756 bp | endogenous_replaced_by_luciferase | TSS cl...
ENST00000870828 · TSS-proximal
2201 bp | endogenous_excluded_vector_luciferase_reta...
ENST00000223095 · genomic 5' UTR
3293 bp | endogenous_replaced_by_luciferase | TSS cl...
ENST00000223095 · spliced 5' UTR
2145 bp | endogenous_replaced_by_luciferase | TSS cl...
ENST00000223095 · TSS-proximal
2201 bp | endogenous_excluded_vector_luciferase_reta...

Ensembl-annotated regulatory regions

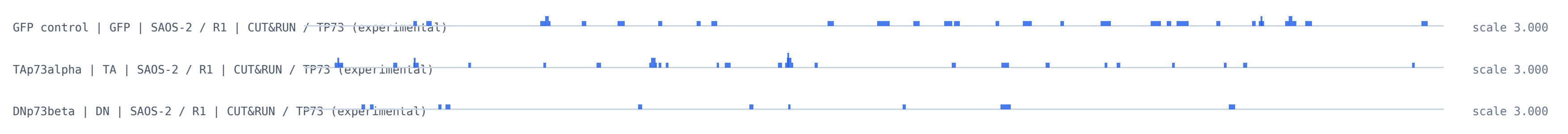
`Ensembl Regulation | release 2026-08 | pipeline 2.1 | GRCh38`


emar | GRCh38:101116944-101117888 | 4270 bp upstream of nearest TSS | no repor
enhancer | GRCh38:101116944-101117888 | 4270 bp upstream of nearest TSS | no r
emar | GRCh38:101118296-101118899 | 3259 bp upstream of nearest TSS | no repor
enhancer | GRCh38:101118296-101118899 | 3259 bp upstream of nearest TSS | no r
emar | GRCh38:101119806-101120597 | 1561 bp upstream of nearest TSS | reporter
enhancer | GRCh38:101119806-101120597 | 1561 bp upstream of nearest TSS | repo
emar | GRCh38:101121249-101121575 | 583 bp upstream of nearest TSS | reporter
enhancer | GRCh38:101121249-101121575 | 583 bp upstream of nearest TSS | repor
promoter | GRCh38:101121823-101122168 | contains TSS | reporter overlap: arch_
emar | GRCh38:101121823-101122982 | contains TSS | reporter overlap: arch_ensg
enhancer | GRCh38:101122169-101122982 | 11 bp downstream of nearest TSS | repo
emar | GRCh38:101124321-101125221 | 1380 bp upstream of nearest TSS | reporter
enhancer | GRCh38:101124321-101125221 | 1380 bp upstream of nearest TSS | repo
emar | GRCh38:101126027-101127627 | contains TSS | reporter overlap: arch_ensg
promoter | GRCh38:101126111-101126611 | contains TSS | reporter overlap: arch_
promoter | GRCh38:101126614-101127129 | contains TSS | reporter overlap: arch_
ctcf | GRCh38:101126823-101126844 | 222 bp downstream of nearest TSS | reporte
enhancer | GRCh38:101127130-101127627 | 26 bp downstream of nearest TSS | repo
emar | GRCh38:101128000-101128958 | 896 bp downstream of nearest TSS | reporte
emar | GRCh38:101129172-101129425 | 2068 bp downstream of nearest TSS | no rep
enhancer | GRCh38:101129172-101129425 | 2068 bp downstream of nearest TSS | no
emar | GRCh38:101130183-101130799 | 3079 bp downstream of nearest TSS | no rep
enhancer | GRCh38:101130183-101130799 | 3079 bp downstream of nearest TSS | no
emar | GRCh38:101131833-101132355 | 4729 bp downstream of nearest TSS | no rep

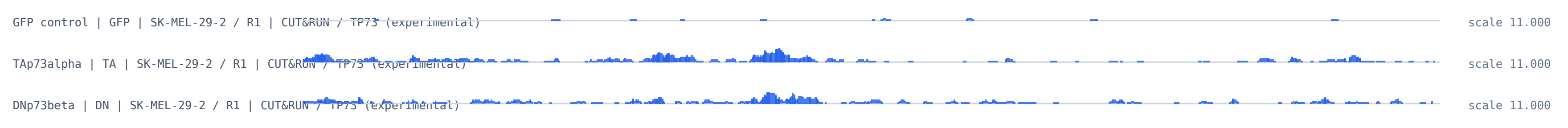
`13 additional source-bound row(s) omitted from SVG; complete rows remain in JSON.`

Projected occupancy grouped by declared biological context

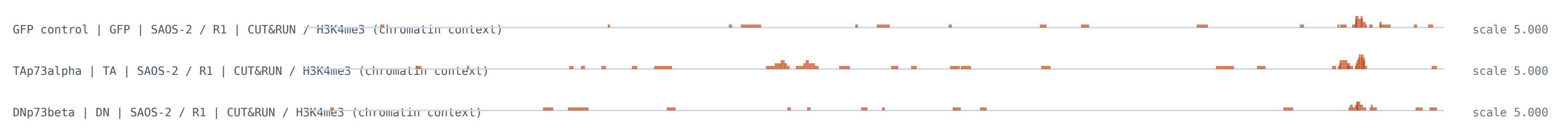
TP73 occupancy — SAOS-2 | shared within group | observed group max 3.000



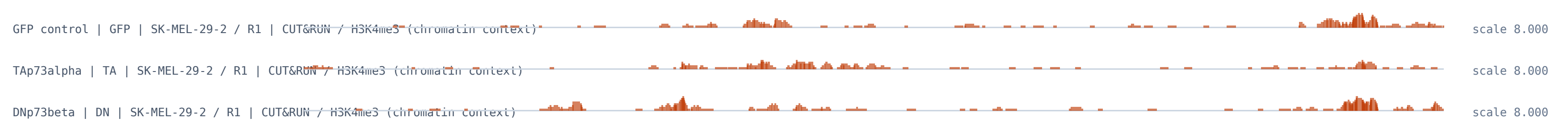
TP73 occupancy — SK-MEL-29-2 | shared within group | observed group max 11.000



H3K4me3 chromatin context — SAOS-2 | shared within group | observed group max 5.000

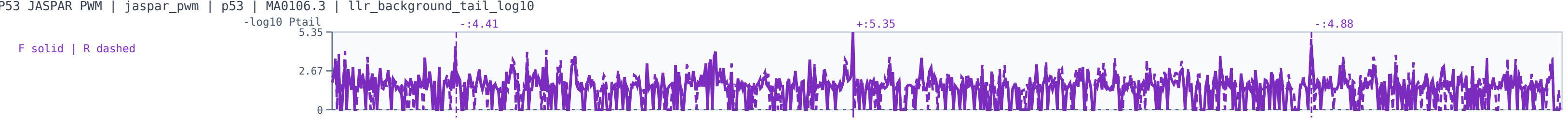


H3K4me3 chromatin context — SK-MEL-29-2 | shared within group | observed group max 8.000



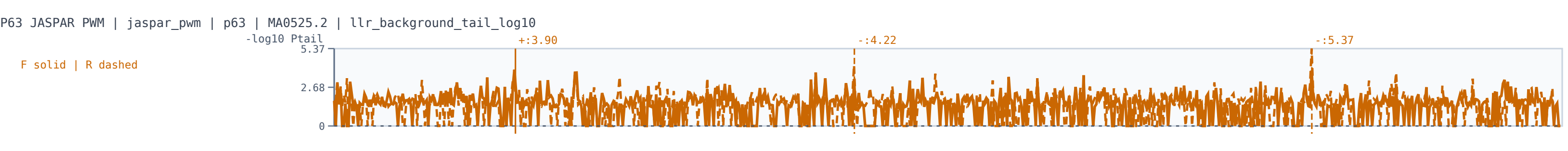
Predicted TF binding scores

`TP53 JASPAR PWM | jaspar_pwm | p53 | MA0106.3 | llr_background_tail_log10`



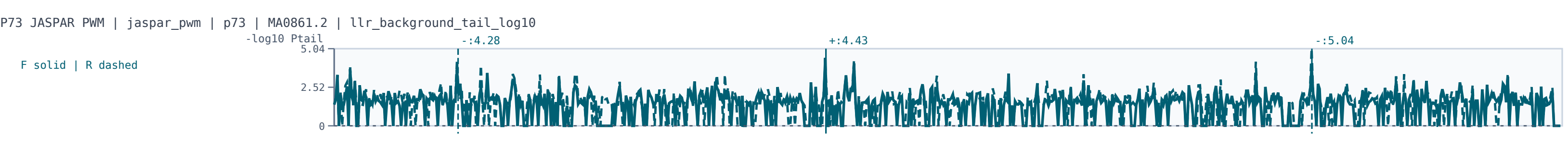
display 0.000..5.345 | -log10 modeled background-tail probability | matrix_spe
JASPAR PWM sequence-match prediction; not measured affinity and not cross-matr

`TP63 JASPAR PWM | jaspar_pwm | p63 | MA0525.2 | llr_background_tail_log10`



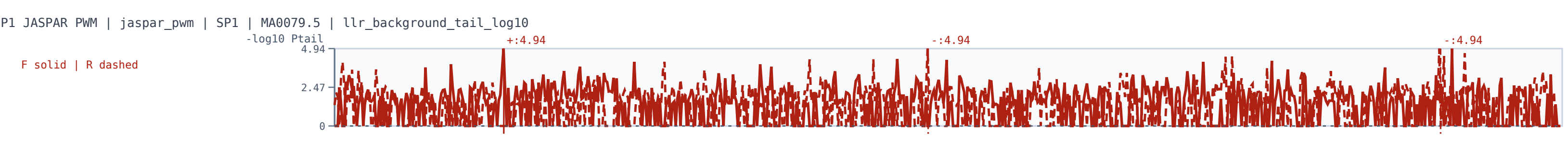
display 0.000..5.370 | -log10 modeled background-tail probability | matrix_spe
JASPAR PWM sequence-match prediction; not measured affinity and not cross-matr

`TP73 JASPAR PWM | jaspar_pwm | p73 | MA0861.2 | llr_background_tail_log10`



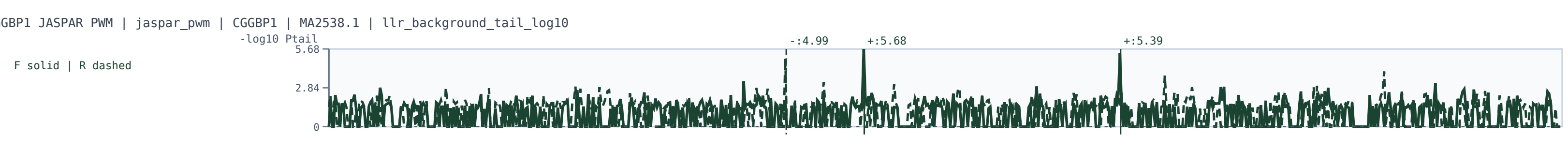
display 0.000..5.037 | -log10 modeled background-tail probability | matrix_spe
JASPAR PWM sequence-match prediction; not measured affinity and not cross-matr

`SP1 JASPAR PWM | jaspar_pwm | SP1 | MA0079.5 | llr_background_tail_log10`



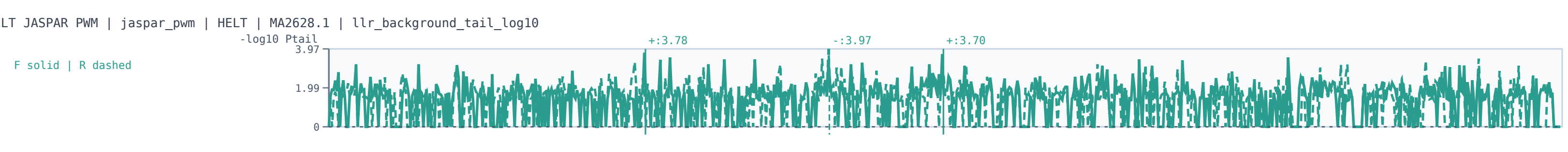
display 0.000..4.941 | -log10 modeled background-tail probability | matrix_spe
JASPAR PWM sequence-match prediction; not measured affinity and not cross-matr

`CGGBP1 JASPAR PWM | jaspar_pwm | CGGBP1 | MA2538.1 | llr_background_tail_log10`



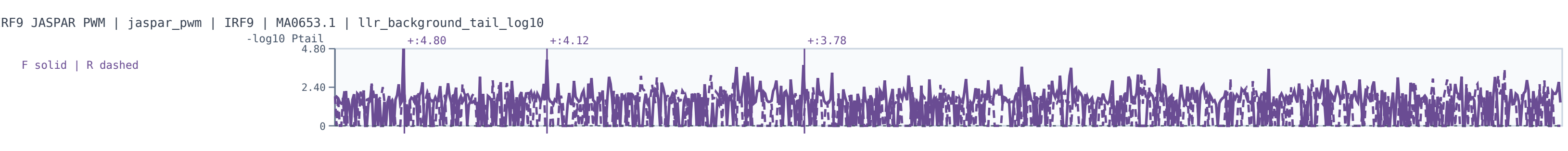
display 0.000..5.681 | -log10 modeled background-tail probability | matrix_spe
JASPAR 2026 CORE matrix-specific PWM score; not calibrated as biochemical affi

`HELT JASPAR PWM | jaspar_pwm | HELT | MA2628.1 | llr_background_tail_log10`



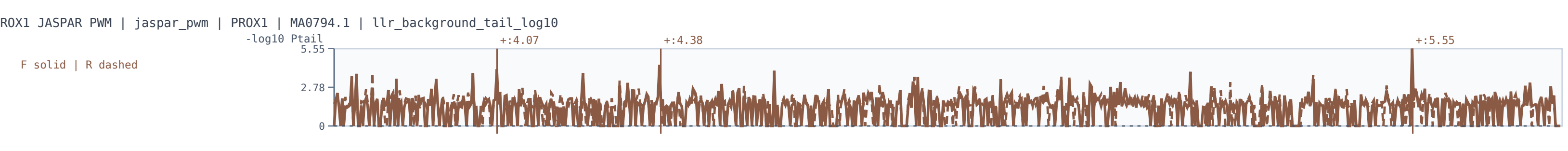
display 0.000..3.971 | -log10 modeled background-tail probability | matrix_spe
JASPAR 2026 CORE matrix-specific PWM score; not calibrated as biochemical affi

`IRF9 JASPAR PWM | jaspar_pwm | IRF9 | MA0653.1 | llr_background_tail_log10`



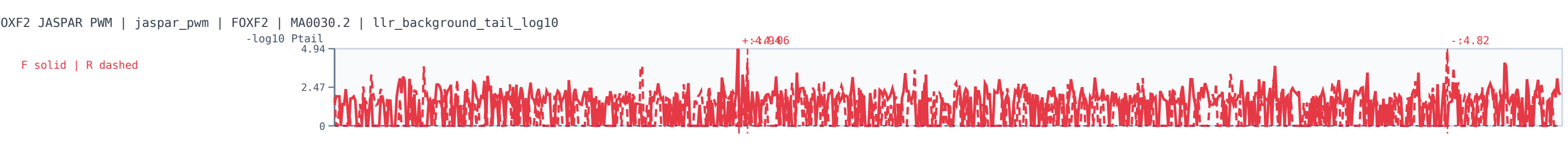
display 0.000..4.801 | -log10 modeled background-tail probability | matrix_spe
JASPAR 2026 CORE matrix-specific PWM score; not calibrated as biochemical affi

`PROX1 JASPAR PWM | jaspar_pwm | PROX1 | MA0794.1 | llr_background_tail_log10`



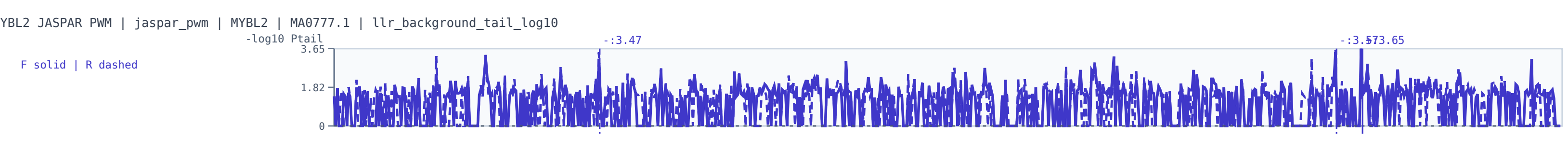
display 0.000..5.553 | -log10 modeled background-tail probability | matrix_spe
JASPAR 2026 CORE matrix-specific PWM score; not calibrated as biochemical affi

`FOXF2 JASPAR PWM | jaspar_pwm | FOXF2 | MA0030.2 | llr_background_tail_log10`



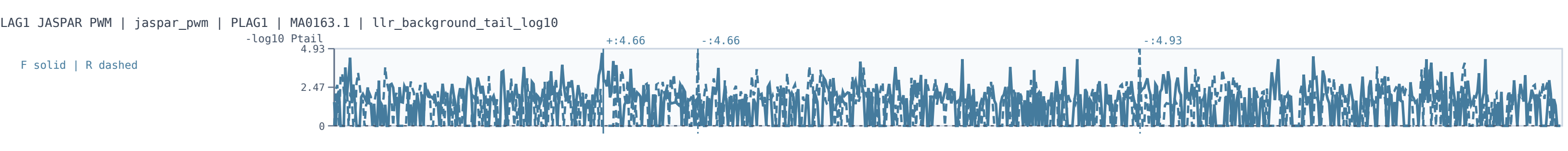
display 0.000..4.941 | -log10 modeled background-tail probability | matrix_spe
JASPAR 2026 CORE matrix-specific PWM score; not calibrated as biochemical affi

`MYBL2 JASPAR PWM | jaspar_pwm | MYBL2 | MA0777.1 | llr_background_tail_log10`



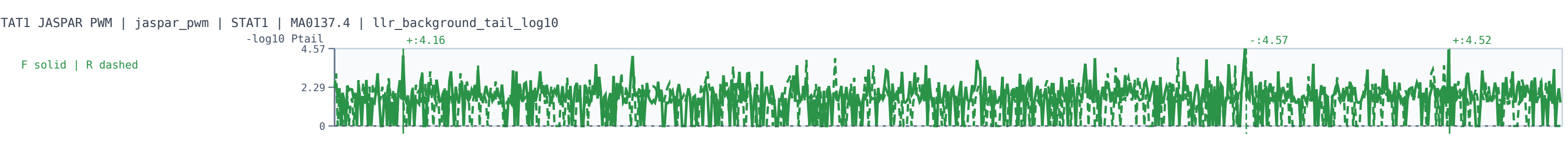
display 0.000..3.646 | -log10 modeled background-tail probability | matrix_spe
JASPAR 2026 CORE matrix-specific PWM score; not calibrated as biochemical affi

`PLAG1 JASPAR PWM | jaspar_pwm | PLAG1 | MA0163.1 | llr_background_tail_log10`



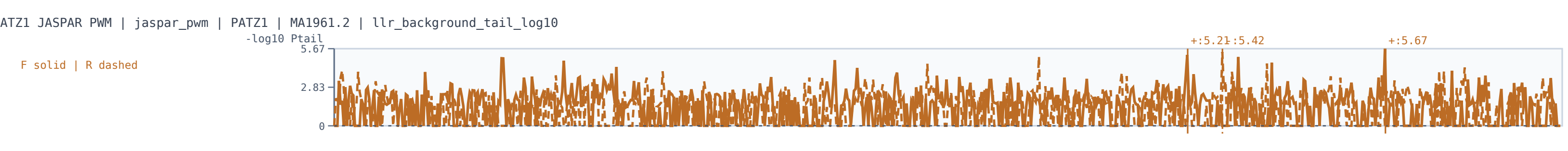
display 0.000..4.935 | -log10 modeled background-tail probability | matrix_spe
JASPAR 2026 CORE matrix-specific PWM score; not calibrated as biochemical affi

`STAT1 JASPAR PWM | jaspar_pwm | STAT1 | MA0137.4 | llr_background_tail_log10`



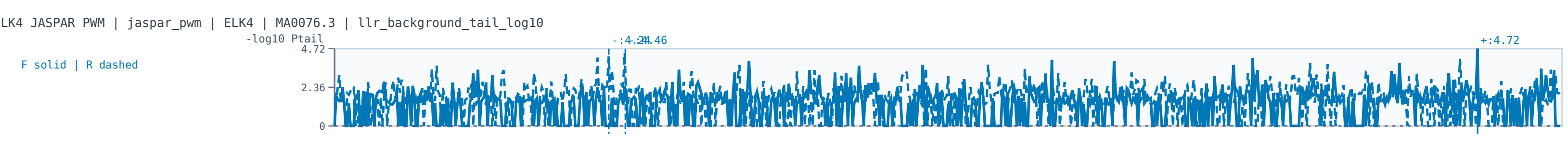
display 0.000..4.573 | -log10 modeled background-tail probability | matrix_spe
JASPAR 2026 CORE matrix-specific PWM score; not calibrated as biochemical affi

`PATZ1 JASPAR PWM | jaspar_pwm | PATZ1 | MA1961.2 | llr_background_tail_log10`



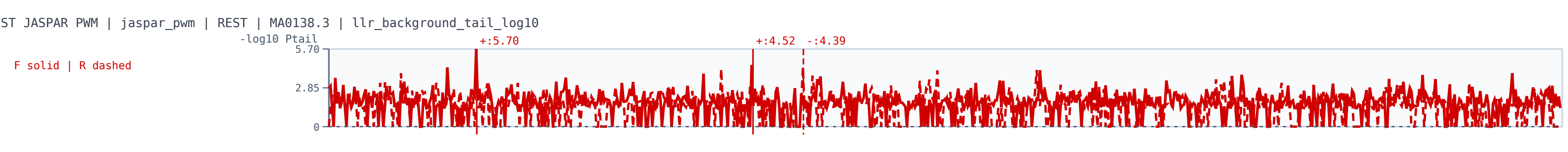
display 0.000..5.668 | -log10 modeled background-tail probability | matrix_spe
JASPAR 2026 CORE matrix-specific PWM score; not calibrated as biochemical affi

`ELK4 JASPAR PWM | jaspar_pwm | ELK4 | MA0076.3 | llr_background_tail_log10`



display 0.000..4.720 | -log10 modeled background-tail probability | matrix_spe
JASPAR 2026 CORE matrix-specific PWM score; not calibrated as biochemical affi

`REST JASPAR PWM | jaspar_pwm | REST | MA0138.3 | llr_background_tail_log10`



display 0.000..5.703 | -log10 modeled background-tail probability | matrix_spe
JASPAR 2026 CORE matrix-specific PWM score; not calibrated as biochemical affi

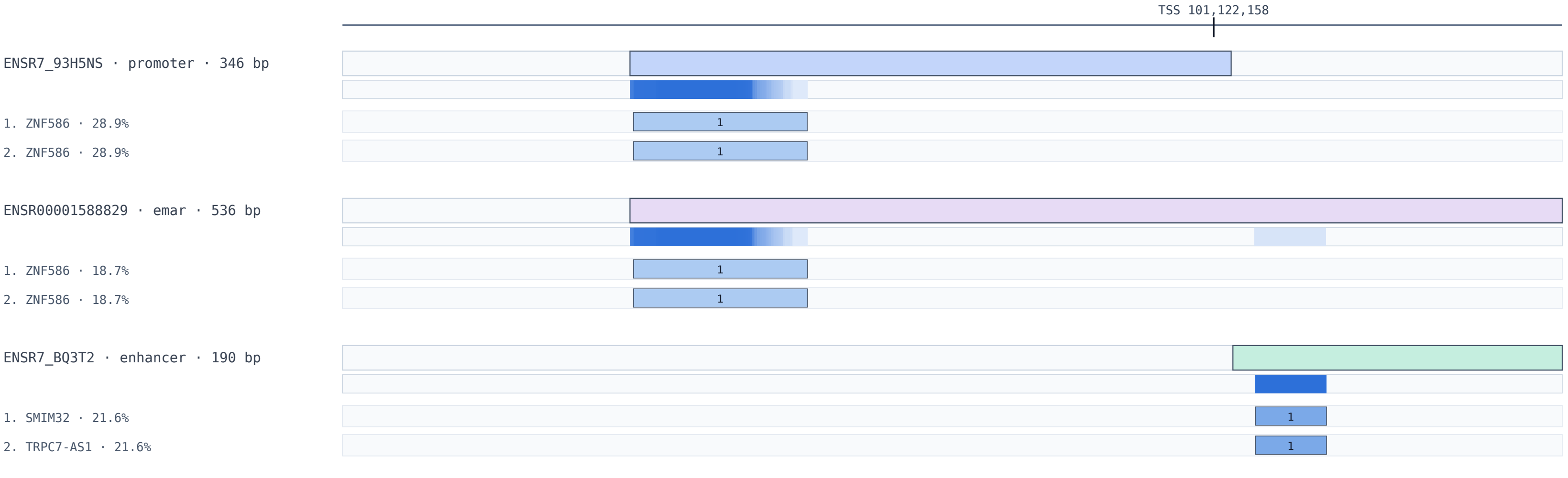
TSS-local Ensembl-feature promoterome similarity

`Transcript-oriented −500/+200 bp; features clipped to the displayed stretch; BLASTN ≥40 bp, ≥80% identity, E≤1e-05; self-locus and same-gene targets excluded.`

Blue frequency strips count distinct other genes by query position. Numbered blocks give target-promoter order; red outlines mark order/orientation breaks.

SERPINE1_tss_stretch_1

`GRCh38 7:101,121,658–101,122,358 (+)`



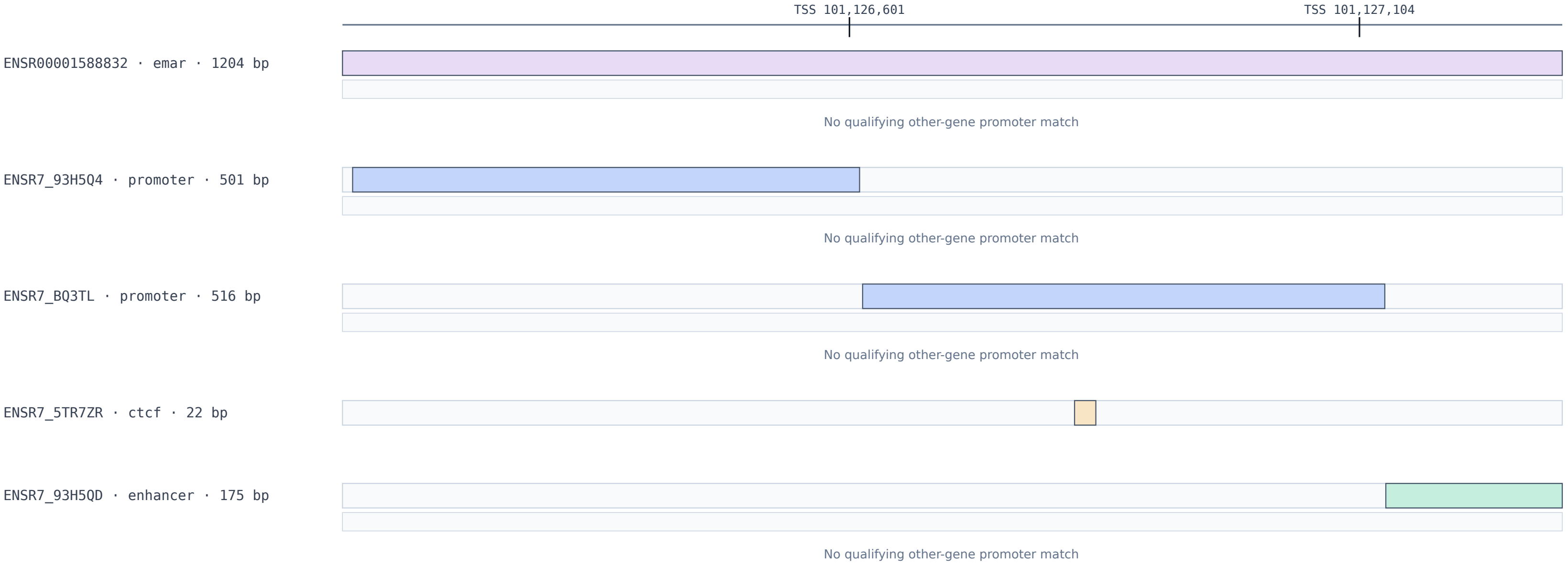
other genes 34,788; ≥25% 12; ≥50% 0; observed; no cap saturation
position frequency; max 34,788 genes

other genes 34,792; ≥25% 0; ≥50% 0; observed; no cap saturation
position frequency; max 34,790 genes

other genes 2; ≥25% 0; ≥50% 0; observed; no cap saturation
position frequency; max 2 genes

SERPINE1_tss_stretch_2

`GRCh38 7:101,126,101–101,127,304 (+)`


TSS 101,126,601
TSS 101,127,104
ENSR00001588832 · emar · 1204 bp
No qualifying other-gene promoter match
ENSR7_93H5Q4 · promoter · 501 bp
No qualifying other-gene promoter match
ENSR7_BQ3TL · promoter · 516 bp
No qualifying other-gene promoter match
ENSR7_5TR7ZR · ctcf · 22 bp
ENSR7_93H5QD · enhancer · 175 bp
No qualifying other-gene promoter match

other genes 0; ≥25% 0; ≥50% 0; observed; no cap saturation
position frequency; max 0 genes

other genes 0; ≥25% 0; ≥50% 0; observed; no cap saturation
position frequency; max 0 genes

other genes 0; ≥25% 0; ≥50% 0; observed; no cap saturation
position frequency; max 0 genes

motif-scale; below 40-bp search gate

other genes 0; ≥25% 0; ≥50% 0; observed; no cap saturation
position frequency; max 0 genes

Similarity is structural evidence; Ensembl classes, predicted TFBS and CUT&RUN enrichment do not prove reporter activity or sufficiency.

Reporter interpretation boundaries

```
Transcript annotation alone does not establish which TSS is used in the assayed cells.
CUT&RUN overlap is occupancy/enrichment evidence, not proof of TSS usage, promoter activity, direct regulation, biochemical affinity, or luciferase output.
Reporter geometry does not verify the identity of a physical laboratory construct.
```

Warnings / interpretation checks

Evidence provenance

```
ensembl_regulation_annotation_index: ensembl_regulation_2026_08_grch38
ensembl_regulation_annotation_intervals: ensembl_regulation_2026_08_grch38
isoform_panel: SERPINE1_panel
occupancy_layout: tp73_saos2,tp73_skmel29_2,h3k4me3_saos2,h3k4me3_skmel29_2
predicted_tf_score_jaspar_pwm: local_jaspar_registry
projected_occupancy_track: E-MTAB-15709:h3k4me3:saos2:DN:R1
projected_occupancy_track: E-MTAB-15709:h3k4me3:saos2:GFP:R1
projected_occupancy_track: E-MTAB-15709:h3k4me3:saos2:TA:R1
projected_occupancy_track: E-MTAB-15709:h3k4me3:skmel29-2:DN:R1
projected_occupancy_track: E-MTAB-15709:h3k4me3:skmel29-2:GFP:R1
```